A quadratic counterexample to the product Hermitian Jacobi eigenfunction conjecture

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Task 1: the original problem. Conjecture 3.12 of [1] asks whether every product polynomial defined by equation (21) is an eigenfunction of the Hermitian Jacobi operator on the matrix simplex (Definition 3.5). We give an interior counterexample for two 2×2 matrices. The result is independent of the nonzero normalization constants used for the Jacobi polynomials.

Theorem 1. *Let $r, s > 0$, $m = k = 2$, and $\kappa = (r, s, s)$. The product polynomial with $\tau_1 = (0, 0)$ and $\tau_2 = (1, 0)$ is not an eigenfunction of $\mathcal{G}_\kappa^2$. In particular, Conjecture 3.12 is false for $\kappa = (1, 1, 1)$, which also belongs to the geometric parameter range $n_1 = n_2 = n_3 = m = 2$.*

Identifying the prescribed polynomial

Write $A = \Lambda_1$, $B = \Lambda_2$, and $S = I_2 - A$. The interior of the matrix simplex is given by $A > 0$, $B > 0$, and $I_2 - A - B > 0$. In equation (21) of [1], the chosen partitions give $a_1 = 2 + 2s$ and $a_2 = s$. The first Jacobi factor has partition zero and is a nonzero constant. The second factor is, up to a nonzero constant,

$$P_{(1,0)}^{(s,s,2)}(Y) = \operatorname{tr} Y - 1.$$

Indeed, this is an invariant polynomial of degree one, orthogonal to constants for the positive beta weight $\det(Y)^{s-1} \det(I_2 - Y)^{s-1}$ on $0 < Y < I_2$. This weight is invariant under $Y \mapsto I_2 - Y$, so its mean trace is 1. This uniquely determines the affine polynomial up to a nonzero scalar. For $s = 1$ the determinant in Definition 1.9 gives the same conclusion directly: the divided difference of $6u^2 - 6u + 1$ at the two eigenvalues is $6(\operatorname{tr} Y - 1)$.

It follows that the specified product polynomial is a nonzero constant multiple of

$$\begin{aligned} F(A, B) &= \det(S)\big(\operatorname{tr}(S^{-1/2} B S^{-1/2}) - 1\big) \\ &= \operatorname{tr}(\operatorname{adj}(S) B) - \det(S) \\ &= \operatorname{tr} A + \operatorname{tr} B - 1 - \det A - M(A, B), \\ M(A, B) &= (\operatorname{tr} A)(\operatorname{tr} B) - \operatorname{tr}(AB). \end{aligned} \tag{1}$$

The last equality uses the elementary 2×2 identities $\operatorname{adj}(I_2 - A) = (1 - \operatorname{tr} A) I_2 + A$ and $\det(I_2 - A) = 1 - \operatorname{tr} A + \det A$. In particular, F is a polynomial in all matrix entries, invariant under simultaneous unitary conjugation.

Applying the full matrix operator

Set $N = r + 2s + 3$, $X_1 = A$, and $X_2 = B$. Definition 3.5 specializes to

$$\begin{aligned}\mathcal{G}f = &\sum_{j=1}^{2}\sum_{a,b,c,d=1}^{2}(I_2 - X_j)_{ad}(X_j)_{cb}\frac{\partial^2 f}{\partial(X_j)_{ab}\partial(X_j)_{cd}}\\ &-\frac{1}{2}\sum_{j\neq\ell}\sum_{a,b,c,d=1}^{2}\big((X_j)_{ad}(X_\ell)_{cb} + (X_j)_{cb}(X_\ell)_{ad}\big)\frac{\partial^2 f}{\partial(X_j)_{ab}\partial(X_\ell)_{cd}}\\ &+\sum_{j=1}^{2}\sum_{a,b=1}^{2}\big((\kappa_j+1)\delta_{ab} - N(X_j)_{ab}\big)\frac{\partial f}{\partial(X_j)_{ab}}.\end{aligned} \tag{2}$$

Derivatives here are taken before restricting to any special matrices. For example, when $A_{12} = u+iv$ and $A_{21} = u-iv$, they mean $\partial_{A_{12}} = (\partial_u - i\partial_v)/2$ and $\partial_{A_{21}} = (\partial_u + i\partial_v)/2$ on the Hermitian space. In particular, setting off-diagonal entries to zero before differentiation would give an incorrect operator calculation.

Direct differentiation in (2) gives

$$\begin{aligned}\mathcal{G}\operatorname{tr}A &= 2(r+1) - N\operatorname{tr}A, & \mathcal{G}\operatorname{tr}B &= 2(s+1) - N\operatorname{tr}B,\\ \mathcal{G}\det A &= r\operatorname{tr}A - 2(N-1)\det A,\\ \mathcal{G}M &= (s+1)\operatorname{tr}A + (r+1)\operatorname{tr}B - 2(N-1)M.\end{aligned} \tag{3}$$

Here the determinant identity follows by differentiating $A_{11}A_{22} - A_{12}A_{21}$. For the last line, the two intermediate identities are

$$\begin{aligned}\mathcal{G}\big((\operatorname{tr}A)(\operatorname{tr}B)\big) &= 2(r+1)\operatorname{tr}B + 2(s+1)\operatorname{tr}A - 2N(\operatorname{tr}A)(\operatorname{tr}B) - 2\operatorname{tr}(AB),\\ \mathcal{G}\operatorname{tr}(AB) &= (r+1)\operatorname{tr}B + (s+1)\operatorname{tr}A - 2N\operatorname{tr}(AB) - 2(\operatorname{tr}A)(\operatorname{tr}B).\end{aligned}$$

Subtracting proves the formula for M. Substitution into (1) now yields the polynomial identity

$$\boxed{\mathcal{G}F = -2(N-1)F + sL, \qquad L(A,B) = \operatorname{tr}A + 2\operatorname{tr}B - 2.} \tag{4}$$

An exact interior witness and the spectral correction

Take

$$A_0 = \begin{pmatrix}1/3 & 0\\ 0 & 2/3\end{pmatrix}, \qquad B_0 = \frac{2}{9}I_2, \qquad I_2 - A_0 - B_0 = \begin{pmatrix}4/9 & 0\\ 0 & 1/9\end{pmatrix}.$$

All three matrices are positive definite. Also $\det(I_2 - A_0) = 2/9$ and $\operatorname{tr}(\operatorname{adj}(I_2 - A_0)B_0) = 2/9$, whence

$$F(A_0, B_0) = 0, \qquad L(A_0, B_0) = -\frac{1}{9}, \qquad (\mathcal{G}F)(A_0, B_0) = -\frac{s}{9} \neq 0.$$

If F were an eigenfunction, $\mathcal{G}F = \lambda F$ for some constant λ, the left side would vanish at every zero of F. This contradiction proves Theorem 1, including the stated case $r = s = 1$ with exact operator value $-1/9$.

The obstruction can also be described explicitly. Equations (3) imply $\mathcal{G}L = -NL$. Since $N > 3$, the polynomial

$$Q = F - \frac{s}{N-2}L \quad \text{satisfies} \quad \mathcal{G}Q = -2(N-1)Q.$$

Thus the proposed product F combines two nonzero eigenfunctions, Q and L, with distinct eigenvalues $-2(N-1)$ and $-N$. An overall rescaling of the operator or of the Jacobi polynomials does not remove the counterexample. No boundary parameter, singular matrix, or omitted regularity assumption is involved.

Verification scope. The preceding identities are the proof. The accompanying exact checker independently differentiates the source operator in all eight formal matrix entries and then in eight real Hermitian coordinates. It also checks the rational witness, the affine Jacobi normalization, and the two-eigenfunction decomposition. No floating-point inference is used. The proof and source audit were performed by the authoring agent; independent peer review is not claimed.

Four follow-up tasks

Task 2. Classify which product indices of total polynomial degree at most four are genuine eigenfunctions for $m = k = 2$, including exceptional parameter sets where distinct eigenvalues coincide.

Task 3. Develop a finite triangular procedure that decomposes each product polynomial into true Jacobi eigenfunctions, and determine whether its coefficients admit formulas in terms of partitions and the parameters.

Task 4. Resolve the separate strong-triangularity Conjecture 3.13, keeping track of multiplicity spaces of the multivariate Schur basis. The present counterexample does not decide that conjecture.

Task 5. For general matrix size m and $k \geq 2$, determine the first product indices that fail the eigenfunction property and construct explicit interior witnesses or prove any surviving families.

References

[1] T. Kuijper, *Jacobi polynomials on the simplex of Hermitian matrices and zonal spherical functions on partial flag manifolds*, arXiv:2607.25760v1 (2026), Definition 1.9, equation (7), Definition 3.5, equation (21), and Conjecture 3.12 (pp. 10–11, 30, 34–35).